**UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS**

**FACULTAD DE INGENIERÍA DE SISTEMAS E INFORMÁTICA**

**DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING TECHNOLOGIES**

# RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED TECHNOLOGIES

**Deliverable:**

Reproducibility Audit Report

**Professor:**

Dr. Loveleen Gaur

**Student:**

MSc. Cendik Leonardo Mauricio Orcón

**2026 - II**

## Executive Summary

**Purpose.** This report applies the Session 6 reproducibility scorecard to the paper "R+R: Towards Reliable and Generalizable Differentially Private Machine Learning" by Bao and Bindschaedler. The paper is an unusually suitable target for this exercise because reproducibility and replicability are themselves central research questions in the study. The authors systematically examine 11 recent state-of-the-art differentially private machine learning (DPML) techniques and investigate whether reported gains can be reproduced and whether they remain valid outside the original experimental conditions.

**Overall judgement.** The audit finds strong evidence of reproducibility practices: extensive seed analysis, explicit statistical testing, effect-size computation, evaluation across multiple datasets and conditions, publicly available source code, and documented computational resources. Three areas remain only partially satisfied under the course scorecard: an explicit end-to-end leakage audit, uniform separation of tuning from final evaluation, and systematic reporting of confidence intervals alongside effect sizes.

| YES | PARTIAL | NO |
|---|---|---|
| 6 | 3 | 0 |

**Operational score: 7.5 / 9 = 83.3% · HIGH REPRODUCIBILITY**

Scoring convention adopted for this report: Yes = 1, Partial = 0.5, No = 0. The Session 6 slide defines the categories but does not prescribe a numeric weighting.

## 1. Audit Method and Evidence Base

The scorecard is taken from the Session 6 course material. Each item was evaluated as Yes, Partial, or No according to whether the paper and its public research artifact provide clear, sufficient, and reproducible evidence. A "Partial" judgement is used when a practice is present but incomplete, unevenly documented, or not demonstrated across the full experimental pipeline.

| Evidence source | Role in the audit | Availability |
|---|---|---|
| ACSAC 2024 paper | Primary peer-reviewed publication; establishes the research questions, methodology, and results. | Public conference record / IEEE paper |
| Extended version | Provides a broader literature overview and additional methodological detail. | Public PDF / arXiv |
| Official GitHub repository | Used to inspect scripts, environment requirements, seed experiments, statistical-test code, and computational instructions. | Public repository |
| ACSAC artifact page | Confirms the paper participated in the conference artifact-availability program. | Public conference page |

## 2. Paper Metadata

| | |
|---|---|
| **Title** | R+R: Towards Reliable and Generalizable Differentially Private Machine Learning |
| **Authors** | Wenxuan Bao; Vincent Bindschaedler |
| **Affiliation** | University of Florida |
| **Venue** | 2024 Annual Computer Security Applications Conference (ACSAC) |
| **Pages** | 666–682 |
| **DOI** | 10.1109/ACSAC63791.2024.00062 |
| **Research area** | Differential privacy; machine learning; reproducibility and replicability |
| **Public artifact** | https://github.com/wenxuan-Bao/Reliable-and-Generalizable-DPML |

**Research focus.** The paper conducts a systematic reproducibility and replicability study of 11 recent DPML techniques. The authors explicitly investigate whether claimed state-of-the-art improvements can be reproduced and whether those improvements continue to hold when tested outside their original experimental conditions. The work also emphasizes a DPML-specific challenge: added privacy noise introduces additional randomness, making results more variable and making single-run reporting particularly risky.

## 3. Experimental Setup: What the Study Actually Does

The study combines reproduction of previously published DPML methods with new experiments designed to test robustness and generalization. The public repository includes multiple imported or adapted implementations corresponding to methods evaluated in the paper, as well as scripts for training, fine-tuning, random-seed analysis, seed-hacking experiments, statistical testing, and hyperparameter selection.

- The repository contains scripts specifically dedicated to random-seed experiments (`random_seed.py` and `random_seed_finetune.py`).
- The README instructs users to train 500 models with different seeds before running the seed-hacking analysis, which provides unusually strong evidence that seed variability was studied systematically.
- The repository includes `st_test.py`, which implements a paired t-test (`scipy.stats.ttest_rel`) and computes Cohen's d.
- The repository includes a `Hyperparam_selection` directory, indicating explicit attention to hyperparameter-search procedures.
- The README identifies datasets used in fine-tuning experiments, including CIFAR-10, EuroSAT, ISIC, PathMNIST, Caltech-256, SUN397, and Oxford-IIIT Pet.
- The environment is partially pinned through Python 3.8, a `requirements.txt` file, and CUDA 11.4.3; the README also documents GPU memory expectations and approximate runtimes.

**Important scope note.** This audit assesses whether the paper and artifact document reproducibility-relevant practices. It does not claim that the entire multi-week GPU workload was independently executed. Therefore, "Yes" means that sufficient evidence is present in the publication and artifact, not that the auditor independently reproduced every numerical result.

# 4. Reproducibility Scorecard

| Checklist item | What to look for | Y | P | N | Audit judgement |
|---|---|---|---|---|---|
| Seeds reported, many runs? | Mean ± SD, not one number | X | | | **YES** |
| Splits described? | Sizes and method; grouped or time-based if needed | X | | | **YES** |
| Leakage checked? | Preprocessing inside folds; no future features | | X | | **PARTIAL** |
| Tuning separated? | Validation set or nested CV | | X | | **PARTIAL** |
| Statistical test named? | Appropriate and paired when comparing models | X | | | **YES** |
| Effect size and CI? | Not only a p-value | | X | | **PARTIAL** |
| External validation? | New dataset, setting, period, site, or population | X | | | **YES** |
| Compute documented? | Hardware, runtime, versions | X | | | **YES** |
| Code and data available? | A stranger could re-run it | X | | | **YES** |

## 4.1 Seeds reported, many runs? — YES

The study explicitly investigates random-seed variability rather than treating it as a nuisance. The public README instructs users to train 500 models from scratch with different seeds before conducting the seed-hacking analysis. The training script defines a reproducibility-oriented seed function that sets PyTorch, CUDA, NumPy, and Python random seeds and configures deterministic cuDNN behavior. This is much stronger than reporting a single favorable run. The paper's central argument also emphasizes that DPML introduces additional randomness through privacy noise, so multiple runs are necessary for reliable conclusions.

*Evidence: `random_seed.py`, `random_seed_finetune.py`, `seeds_hacking.py`, and the GitHub README.*

## 4.2 Splits described? — YES

The study uses established computer-vision benchmark datasets and canonical training/test partitions, and the code explicitly loads separate training and test splits for datasets such as CIFAR-10 and MNIST. The paper and artifact also identify the datasets used in the fine-tuning experiments. For the purposes of this scorecard, the data partitioning is sufficiently described to understand which samples are used for training and evaluation, although some dataset-specific details are delegated to the corresponding original repositories or dataset definitions.

*Evidence: dataset loaders in the public code, paper experimental sections, and the GitHub README.*

## 4.3 Leakage checked? — PARTIAL

The study clearly separates training and test data and evaluates methods on public benchmark datasets. However, it does not provide a single, explicit leakage-audit protocol that systematically traces every preprocessing step, augmentation operation, pretraining source, feature-selection decision, and tuning choice across all 11 reproduced techniques. Absence of such a section does not demonstrate that leakage occurred; it means that the artifact does not fully document leakage prevention at the level expected by this scorecard.

*Audit interpretation: risk appears controlled in many experiments, but explicit end-to-end leakage documentation is incomplete.*

### 4.4 Tuning separated? — PARTIAL

Hyperparameter selection is clearly part of the experimental workflow: the repository contains a dedicated `Hyperparam_selection` directory and multiple configuration files. Nevertheless, a uniform independent validation strategy or nested cross-validation procedure is not documented for every reproduced method. Because the study integrates methods from heterogeneous original codebases, the tuning protocol is not fully standardized across all experiments. This prevents an unqualified "Yes."

*Evidence: `Hyperparam_selection/` in the repository and the paper's discussion of heterogeneous experimental practices.*

### 4.5 Statistical test named? — YES

The artifact explicitly names and implements a paired t-test using `scipy.stats.ttest_rel`. The paired design is appropriate when comparing two methods on matched results generated under corresponding experimental conditions. The code reports the t-statistic and p-value, so the inferential procedure is transparent and directly inspectable.

*Evidence: `st_test.py` in the official GitHub repository.*

### 4.6 Effect size and confidence interval? — PARTIAL

The statistical-testing script goes beyond p-values by calculating Cohen's d from the paired result arrays, which is a strong practice because it quantifies the magnitude of the difference. However, the public statistical script does not compute confidence intervals, and confidence intervals are not systematically reported across the principal comparisons. The criterion therefore receives "Partial": the effect-size requirement is satisfied, but the confidence-interval requirement is not consistently satisfied.

*Evidence: `st_test.py` computes a paired t-test and Cohen's d but no confidence interval.*

### 4.7 External validation? — YES

A core research question is whether improvements continue to hold outside the experimental conditions of the original papers. The authors deliberately evaluate techniques across additional datasets, architectures, privacy settings, and fine-tuning configurations. In this methodological context, that functions as external validation of the claimed methodological improvements: the study tests whether a technique generalizes beyond the narrow setting in which it was originally reported.

*Evidence: the ACSAC abstract explicitly states that some methods falter when tested outside their initial experimental conditions; the repository exposes experiments over multiple datasets and settings.*

### 4.8 Compute documented? — YES

The repository documents the computational environment in meaningful detail. It specifies a Python 3.8 environment, installation through `requirements.txt`, CUDA 11.4.3, a recommendation of at least 16 GB of GPU memory, and the actual use of a single NVIDIA A100 GPU with 80 GB of memory. The README also reports that most experiments should finish within 24 hours on one GPU, whereas training 500 seed variants may require approximately one to two weeks.

*Evidence: "Installation" and "Computational Resources and time" sections of the GitHub README.*

### 4.9 Code and data available? — YES

The research code is publicly available and includes the paper PDF, `requirements.txt`, training scripts, statistical-test code, random-seed experiments, hyperparameter-selection code, and snapshots or adaptations of several methods evaluated in the study. The datasets are established public benchmarks and are loaded programmatically. The README notes that reproducing some

constituent methods may require consulting their original repositories or creating separate environments, so the artifact is not fully one-click reproducible; nevertheless, the research objects needed to attempt reproduction are publicly accessible. ACSAC also lists this work in its artifact program.

*Evidence: official GitHub repository and ACSAC artifact listing.*

## 5. Statistical Rigor Assessment

The paper is stronger than many machine-learning studies on statistical rigor because it explicitly treats random seeds as a source of experimental uncertainty and provides code for paired statistical testing. The repository's `st_test.py` computes:

- the mean and sample standard deviation for each result array;
- a paired t-test with a t-statistic and p-value;
- Cohen's d as a standardized effect-size measure.

This combination supports both statistical and practical interpretation. However, the scorecard's requirement is stricter: it asks for effect size and confidence intervals, not merely a p-value. Because confidence intervals are not systematically generated by the provided testing script, the report marks that item as Partial.

## 6. Data Leakage Assessment

No direct evidence of deliberate or obvious leakage was identified in the inspected artifact. The code uses separate dataset-provided training and test partitions, and the study is explicitly concerned with reliable evaluation. Nevertheless, a reproducibility audit distinguishes between "no evidence of leakage" and "an explicit leakage-control protocol is documented." The latter is not uniformly present across all reproduced methods.

A stronger leakage section would document, for every experiment, whether any transform or statistic is learned from data, how preprocessing is fitted, whether pretraining data overlap with evaluation datasets, how augmentation is applied, and whether test performance is ever used to make training or model-selection decisions. This is the basis for the Partial rating.

## 7. Missing or Insufficiently Documented Information

- A single experiment-by-experiment leakage checklist is not provided for all reproduced methods.
- A uniform validation or nested-cross-validation protocol for hyperparameter tuning is not documented across the entire study.
- Confidence intervals are not systematically reported together with effect sizes and p-values.
- Some reproduced methods depend on upstream repositories and potentially different environments, so the public artifact is not a single one-command workflow.
- The public repository documents major compute requirements well, but complete experiment provenance (for example, one manifest mapping every table result to a precise commit/configuration) could be stronger.

## 8. Overall Reproducibility Score

For a concise summary, this report adopts the following operational scoring convention:

| Rating | Points |
|---|---|

| | |
|---|---|
| Yes | 1.0 |
| Partial | 0.5 |
| No | 0.0 |

**6 Yes × 1.0 + 3 Partial × 0.5 + 0 No × 0 = 7.5 / 9 = 83.3%**

### Overall classification: HIGH REPRODUCIBILITY

This numerical interpretation is created for this report. The course slide itself does not define a numerical grading formula.

## 9. Overall Assessment

The paper demonstrates a strong commitment to reproducible and reliable machine-learning research. Its major strengths are the explicit study of seed variability, the availability of extensive public code, direct statistical testing, effect-size computation, evaluation outside original experimental settings, and unusually clear documentation of compute resources. These practices directly address several weaknesses highlighted in Session 6: single-run reporting, unexamined randomness, unsupported "state-of-the-art" claims, and insufficient documentation.

The audit nevertheless identifies three areas that prevent a perfect score. First, the paper does not present one systematic leakage-control audit covering every experimental pipeline. Second, the separation between hyperparameter tuning and final evaluation is not uniformly documented across all constituent methods. Third, Cohen's d is available, but confidence intervals are not systematically reported. These are not fatal flaws; rather, they are concrete opportunities to make an already reproducibility-oriented study even more transparent.

## 10. Three Priority Recommendations

### Recommendation 1 — Add an explicit leakage-control protocol

For every experiment, document which transformations, augmentations, feature-selection steps, pretrained resources, and preprocessing statistics can learn from data. State exactly where each step is fitted and verify that no test information is used during training, tuning, early stopping, or model selection.

### Recommendation 2 — Separate tuning from final evaluation

Use and document an independent validation strategy for hyperparameter and model selection. When data are limited, nested cross-validation can separate the inner selection process from the outer performance estimate. The final test set should remain untouched until the model-selection process is complete.

### Recommendation 3 — Add confidence intervals

Extend the existing statistical framework by reporting 95% confidence intervals for performance differences and, where appropriate, for effect sizes. The study already reports variability, paired tests, and Cohen's d; confidence intervals would make uncertainty easier to interpret and align the analysis more closely with the Session 6 reporting guidance.

## 11. Conclusion

The reproducibility audit indicates that "R+R: Towards Reliable and Generalizable Differentially Private Machine Learning" is a strong example of reproducibility-oriented ML research. The study

does more than attempt to reproduce headline accuracy values: it asks whether the claimed improvements survive new seeds, datasets, settings, and methodological scrutiny. The open repository provides concrete evidence for random-seed experiments, statistical testing, effect-size analysis, training workflows, environment requirements, and computational resources.

At the same time, the exercise illustrates a central lesson from Session 6: reproducibility is not an all-or-nothing label. Even a paper dedicated to reproducibility can improve the transparency of leakage prevention, the separation of tuning from final evaluation, and uncertainty reporting. Under the operational score used in this report, the study receives 7.5 out of 9 points (83.3%), classified as high reproducibility.

## References and Public Artifacts


Bao, W., & Bindschaedler, V. (2024). R+R: Towards Reliable and Generalizable Differentially Private Machine Learning. In 2024 Annual Computer Security Applications Conference (ACSAC), 666–682. IEEE. DOI: 10.1109/ACSAC63791.2024.00062. https://doi.org/10.1109/ACSAC63791.2024.00062

Bao, W., & Bindschaedler, V. (2025). Towards Reliable and Generalizable Differentially Private Machine Learning (Extended Version). arXiv:2508.15141. https://arxiv.org/abs/2508.15141

Bao, W., & Bindschaedler, V. Reliable-and-Generalizable-DPML [Source code]. GitHub. https://github.com/wenxuan-Bao/Reliable-and-Generalizable-DPML

Annual Computer Security Applications Conference. (2024). R+R: Towards Reliable and Generalizable Differentially Private Machine Learning — program record. https://www.acsac.org/2024/program/final/s114.html

Annual Computer Security Applications Conference. (2024). Artifact Availability Program. https://www.acsac.org/2024/program/artifacts/

# Appendix A. Evidence Map to Public Repository

| Artifact | Reproducibility evidence | Scorecard relevance |
|---|---|---|
| README.md | Environment, usage instructions, datasets, 500-seed workflow, GPU and runtime information | Seeds; compute; code availability |
| random_seed.py | Sets Python/NumPy/PyTorch/CUDA seeds and deterministic cuDNN behavior; trains random-seed variants | Seeds / repeatability |
| random_seed_finetune.py | Random-seed experiments for fine-tuning workflows | Seeds / many runs |
| seeds_hacking.py | Analyzes the effect of selecting favorable seeds | Randomness / scientific integrity |
| st_test.py | Paired t-test, p-value, pooled SD, Cohen's d | Statistical test; effect size |
| Hyperparam_selection/ | Dedicated hyperparameter-selection implementation | Tuning separation |
| requirements.txt | Environment dependencies | Compute / environment |
| DP-Mix/, DP-RandP/, private_CNN/, src/, etc. | Code snapshots/adaptations for reproduced methods | Code availability / reproducibility |